HSLU Lucerne University of Applied Sciences and Arts

Computer Vision & AI

SW 05

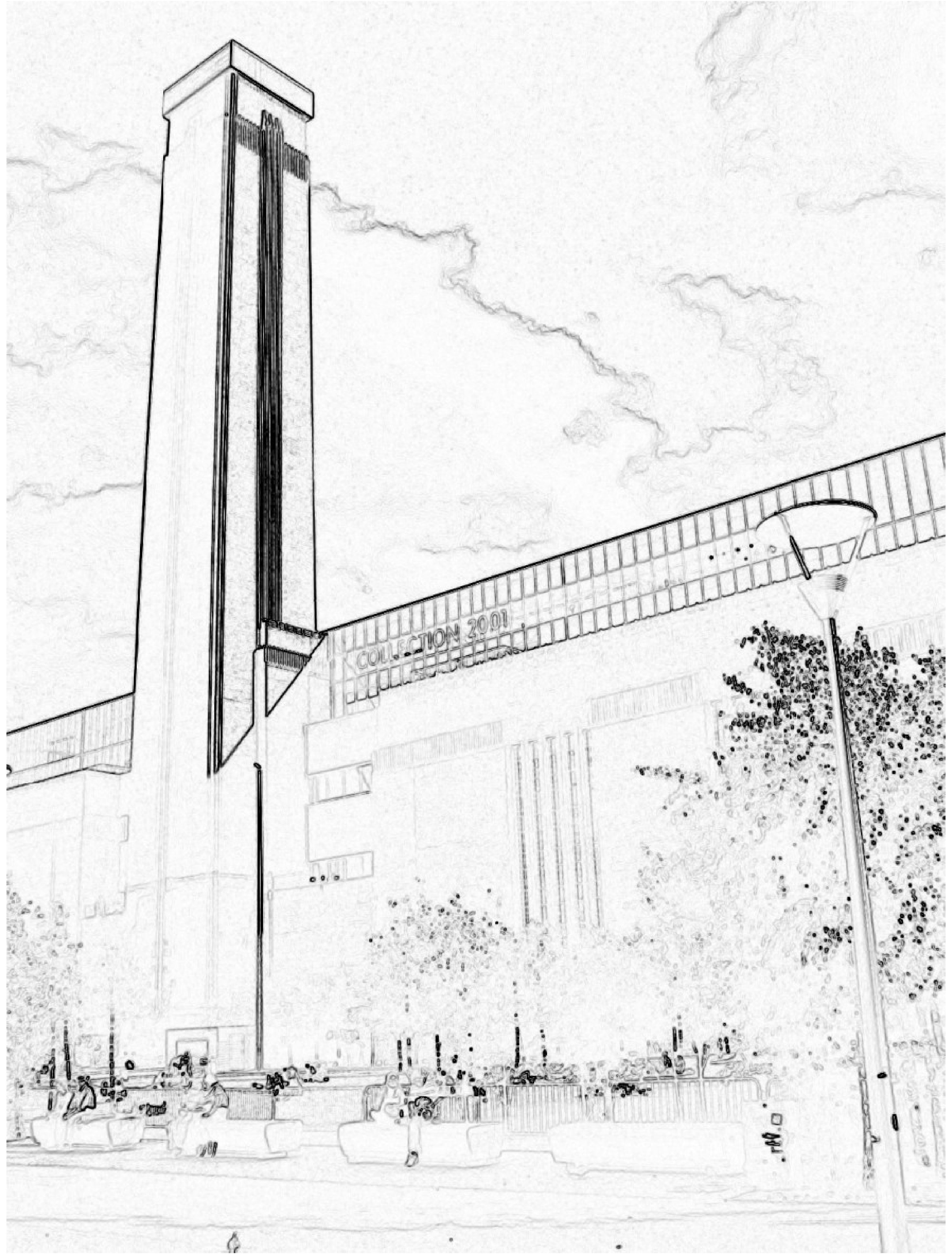
Filter for Edge Detection
Canny Filter

9. Oktober 2024

Segmentation

- object detection
- approaches:
  - boundary based methods
    objects delimited by edges
  - region based methods
    objects determined by their interior
    (e. g. color, texture)
- prerequisite for boundary based segmention
  edge detection

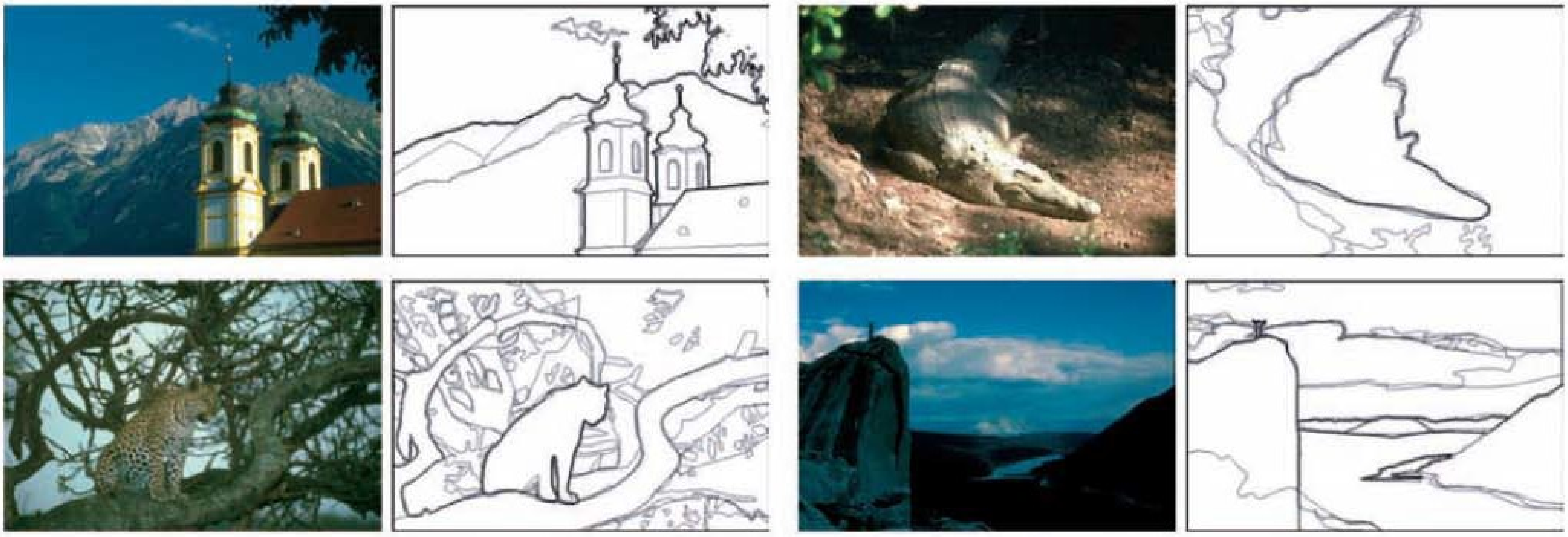
Edge Detection Example



Where do Edges Come From?

- surface normals change direction
- colors adjoin
- change of depth
- change of illumination

Perception of Edges in an Image



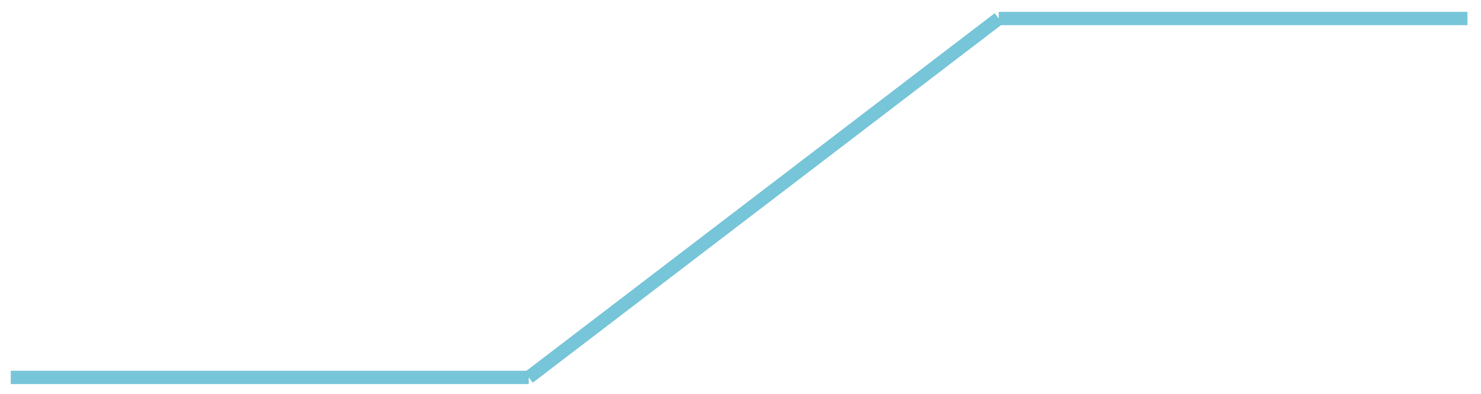
thickness of edges represent where probands draw edges (IEEE 2004, Martin, Fowlkes, and Malik)

1D Edge Detection

Edges are strong changes in image intensity.

Step:

Ramp:



How Can Edges be Found?

Different approaches for edge detection

- Differentiation of image intensity
- Optimal filter fullfilling quality criteria
  (example: Canny Filter)
- Matched filtering
  (optimizing sigal-noise-ratio)

various derivations lead to similar methods

Edge Detection by Differentiation

Numerical computation of first derivative

$$\frac{\partial I}{\partial x} \approx I(x+1, y) - I(x, y)$$

$$\frac{\partial I}{\partial x} \approx I(x+1, y) - I(x-1, y)$$

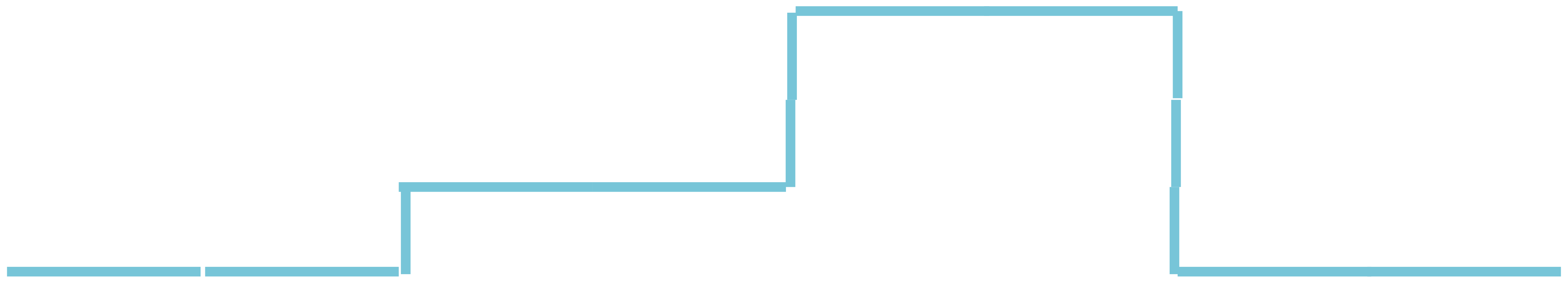
Equivalent to application of filter

-1	**1**

-½	**0**	**½**

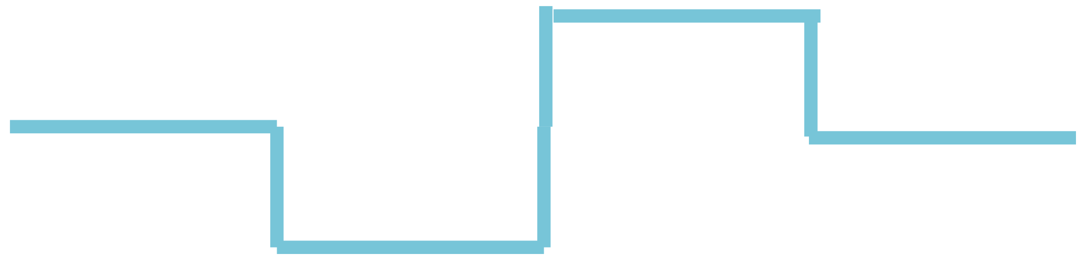
Edge Detection by Differentiation

Image



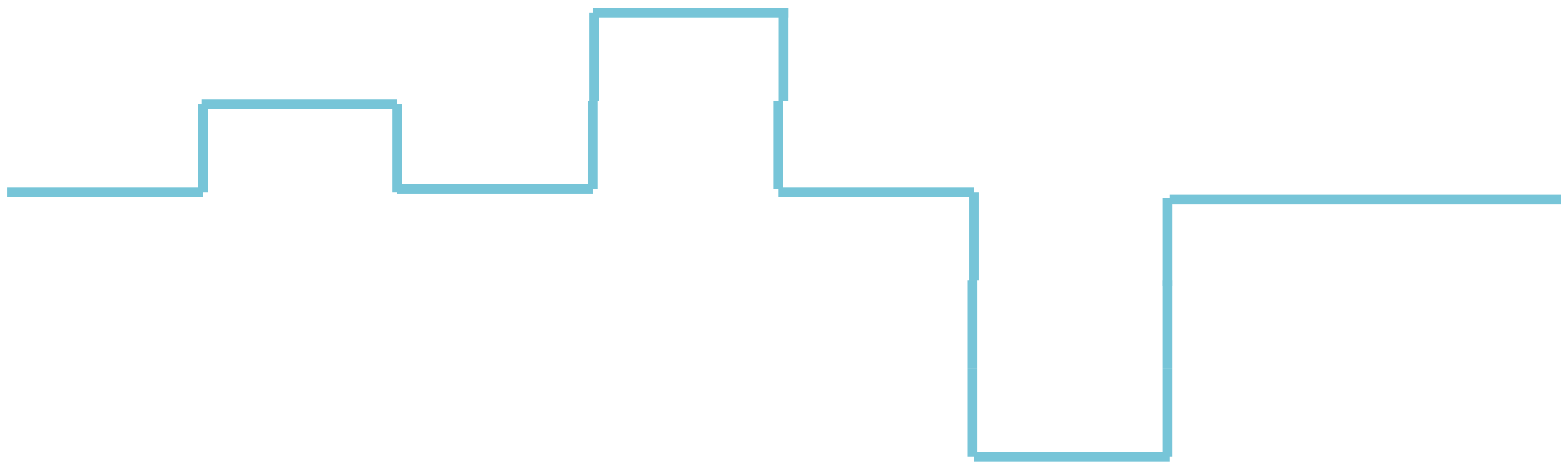
0	0	1	1	3	3	0	0

Filter (correlation)



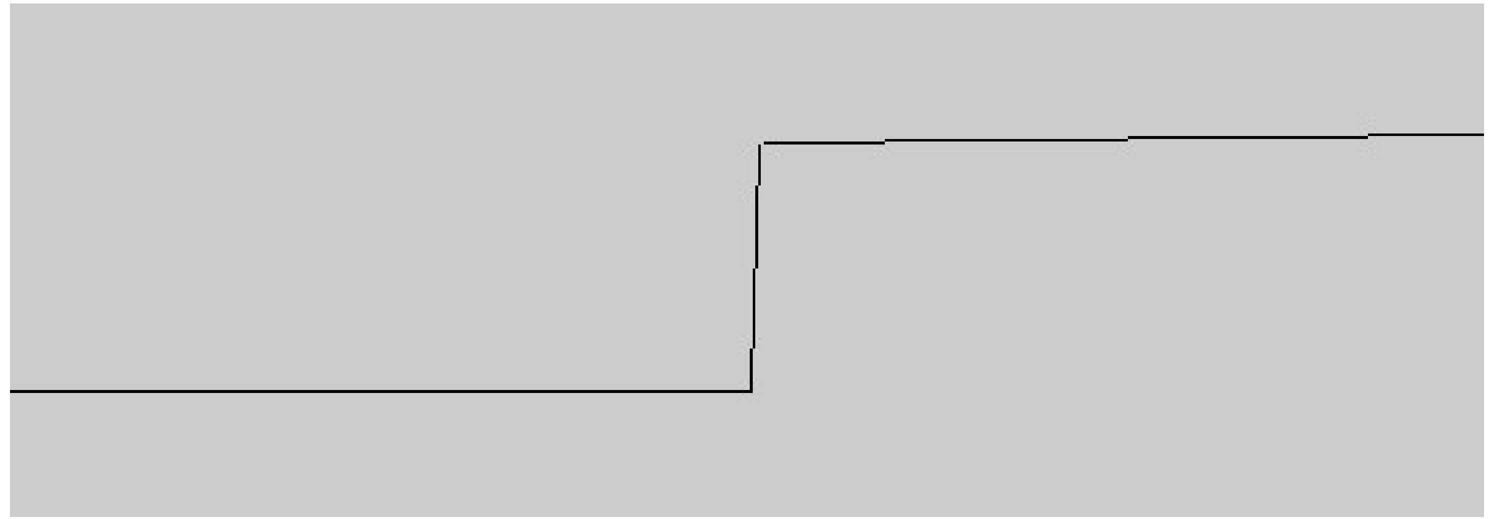
0	-1	1	0

Result

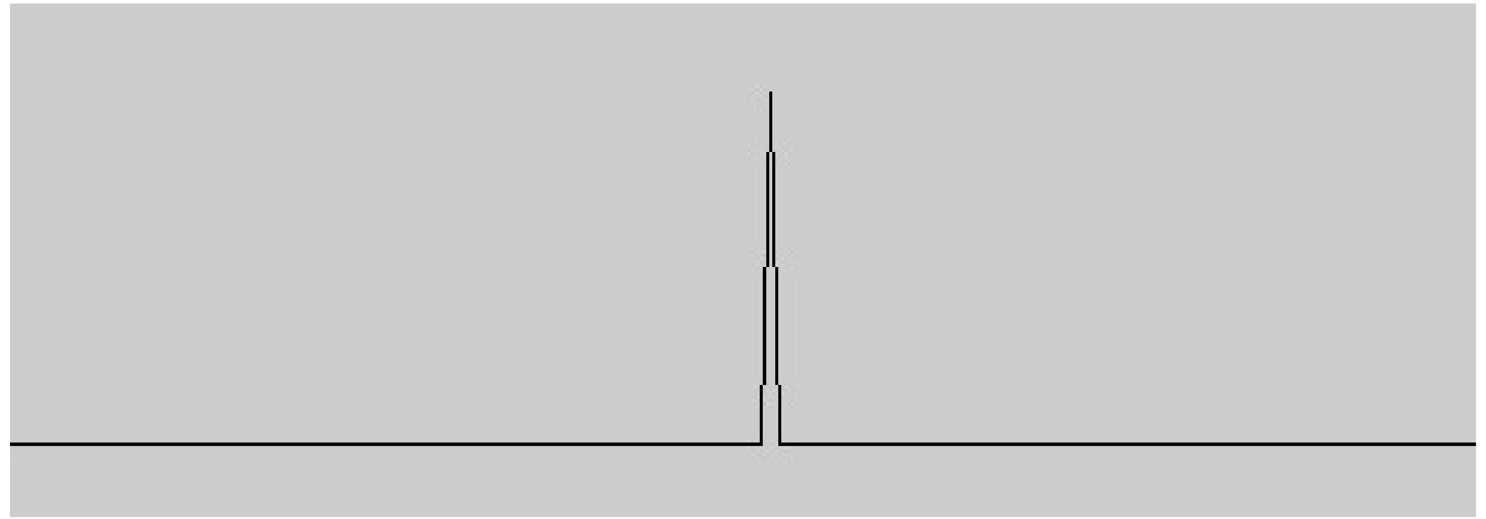


0	1	0	2	0	-3	0	0

Edge Detection by Differentiation, Noise



image



edge

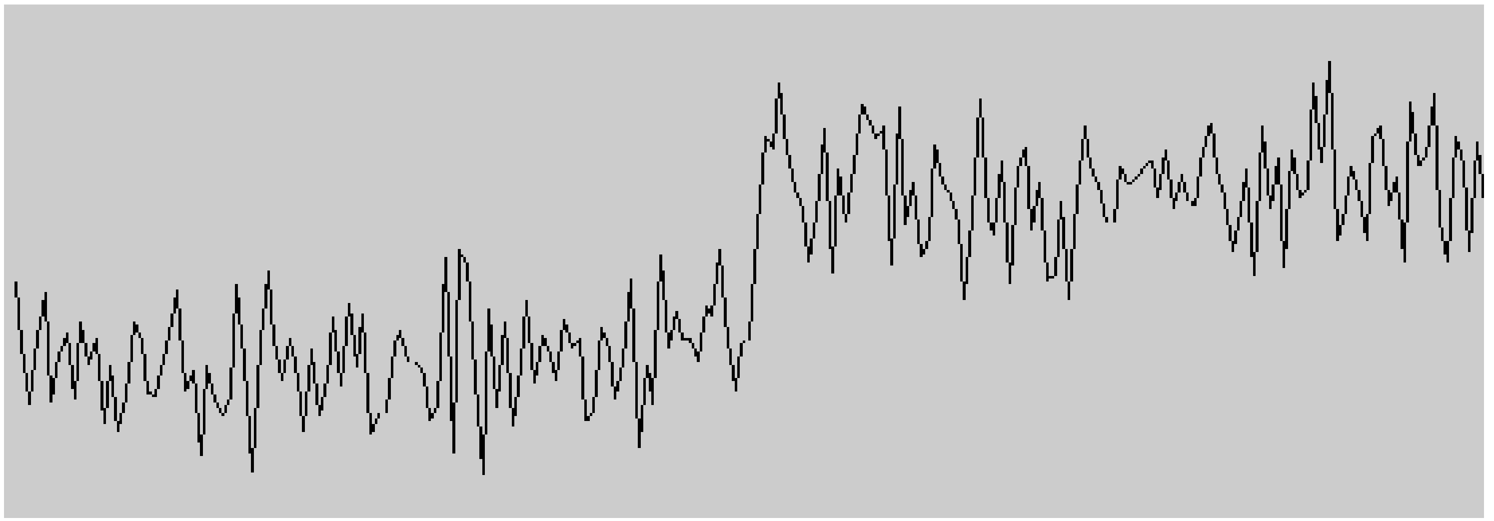
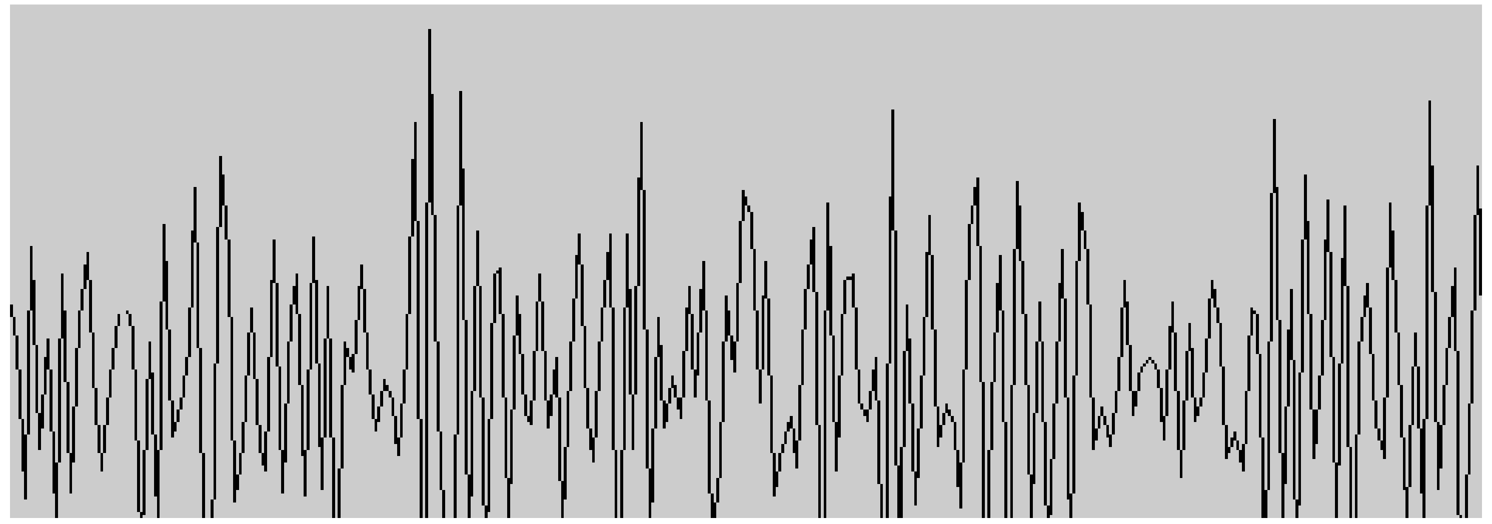


image with noise



problem: many edges

Canny Filter

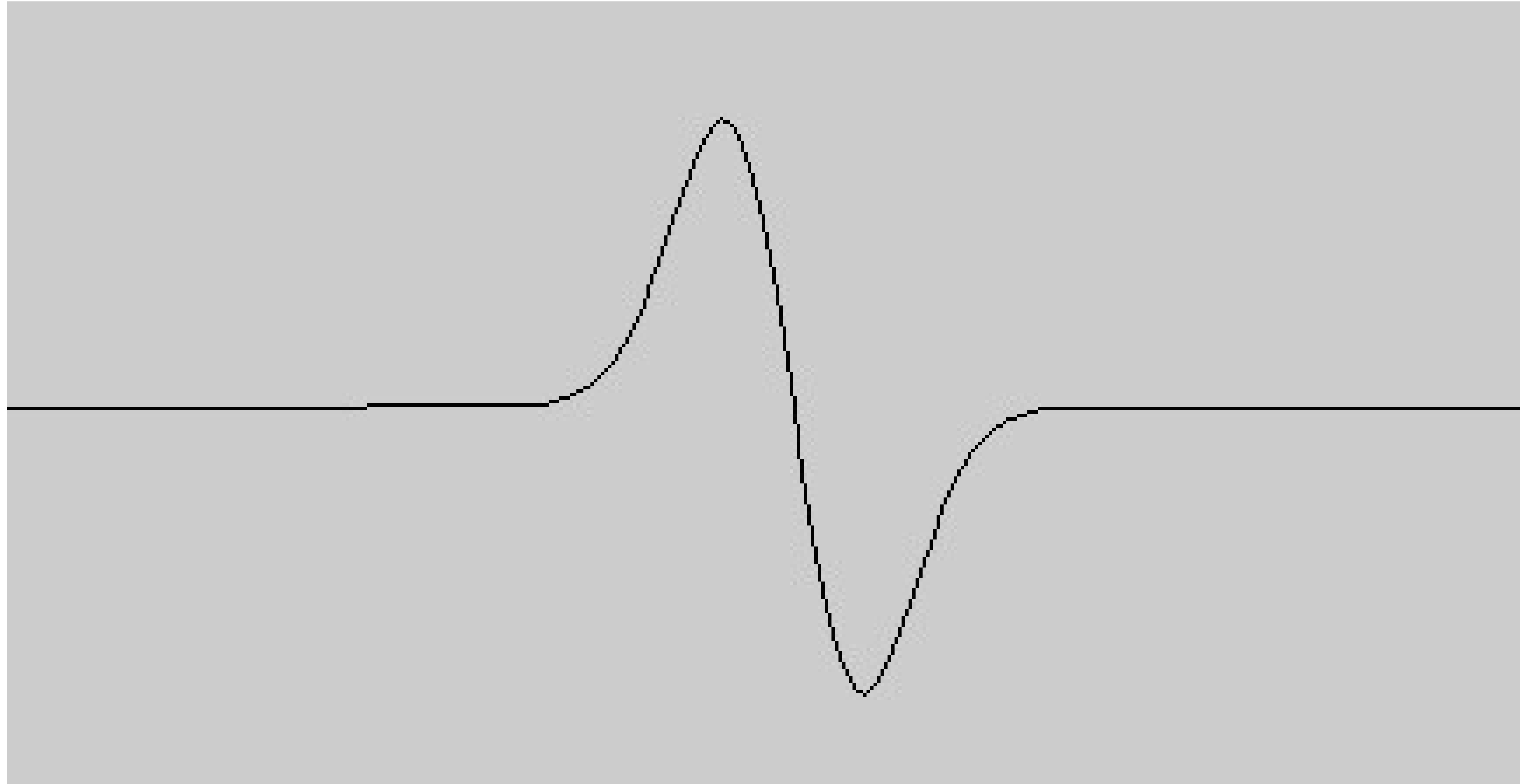
Approach:

Edge detection fullfilling quality criteria

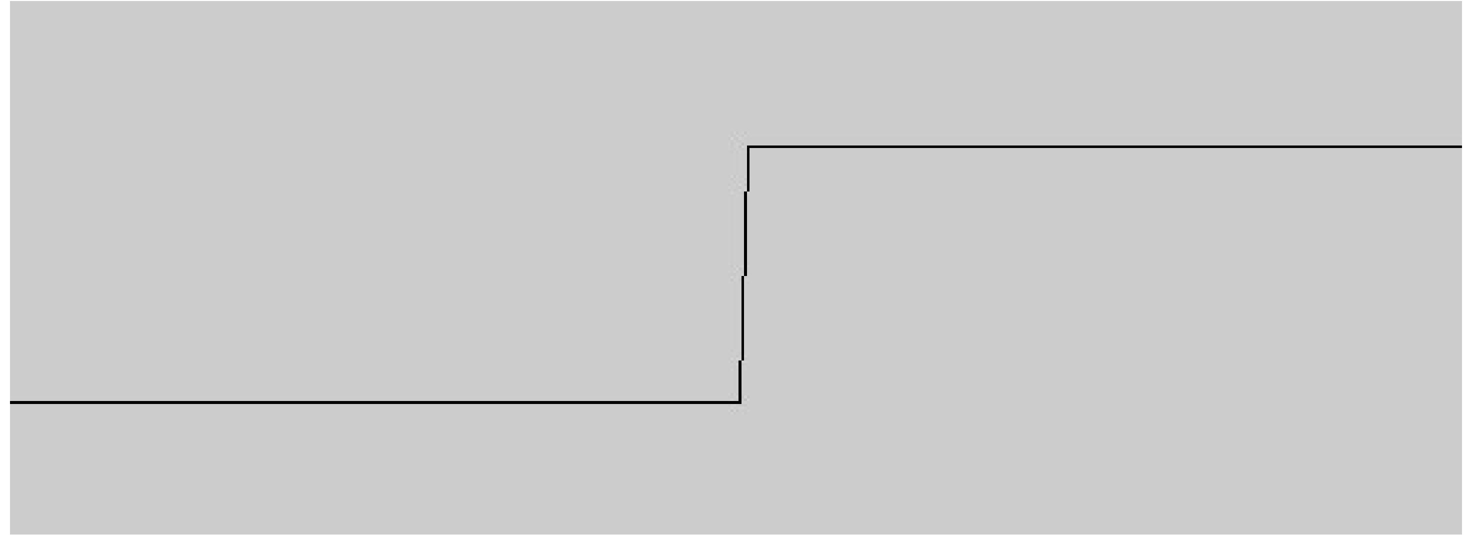
- detection
  strong response of filter at edge
- localization
  computed edge near original edge
- uniqueness
  no other edges found near edge

Canny Filter

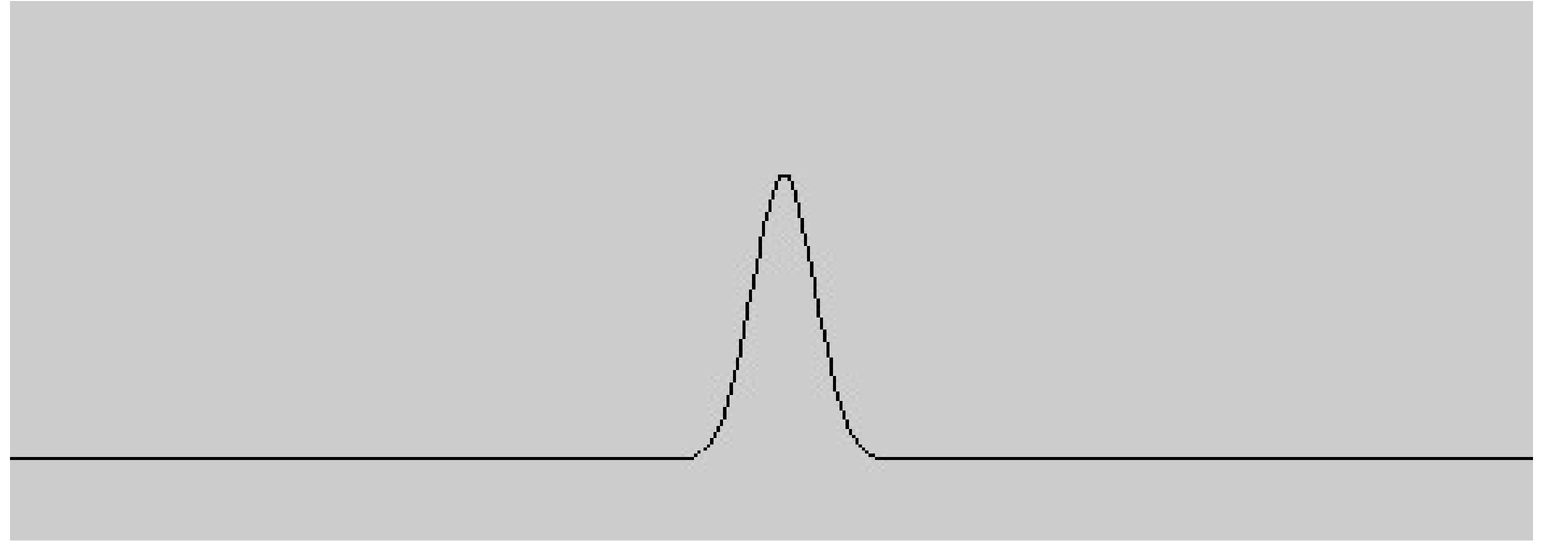
- express quality criteria mathematically
- derive filter from quality criteria analytically
- result: first derivative of Gauss filter is an optimal edge filter (convolution)



Canny Filter



image



edge

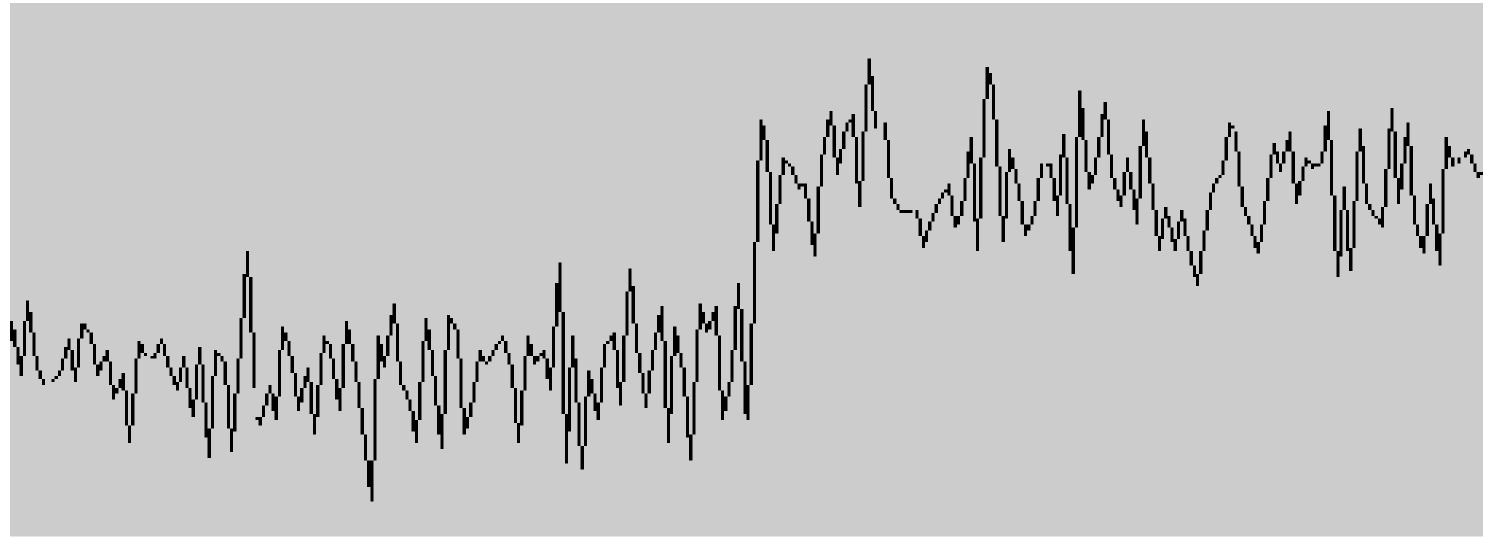
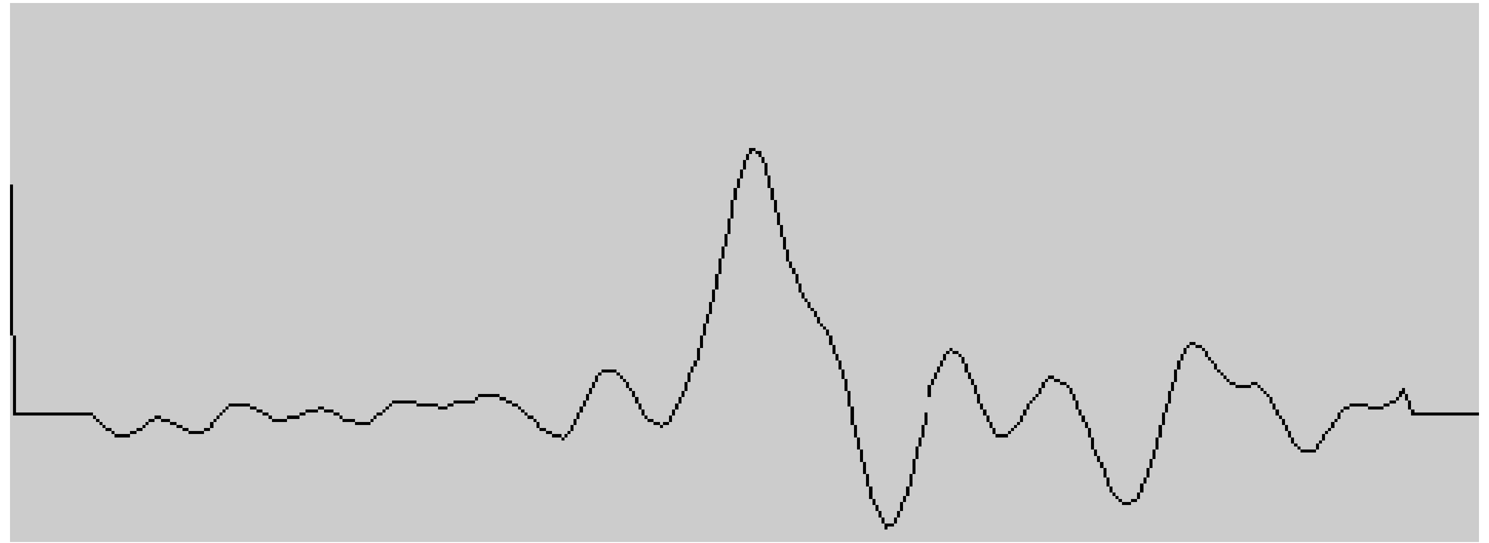


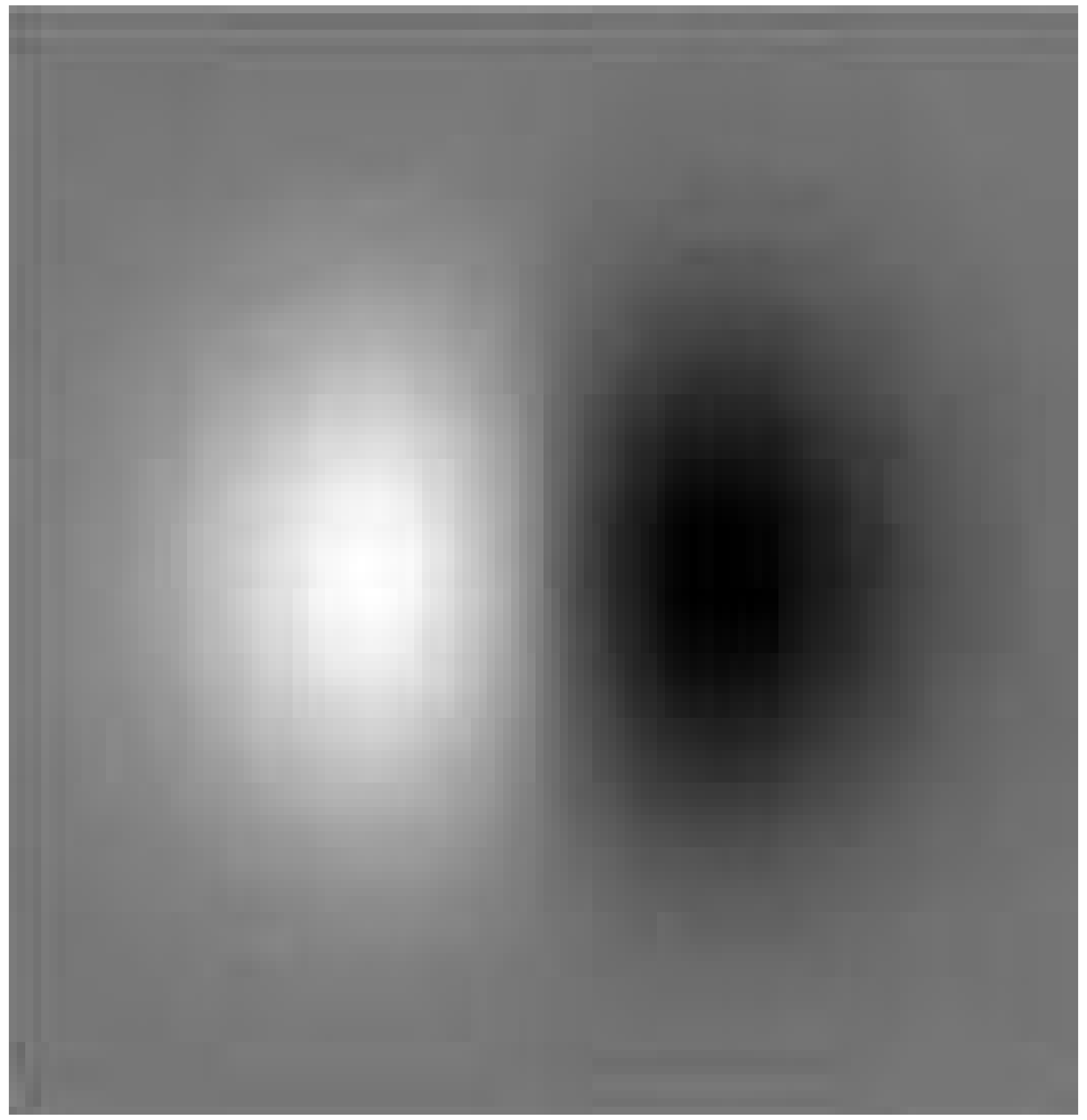
image with noise



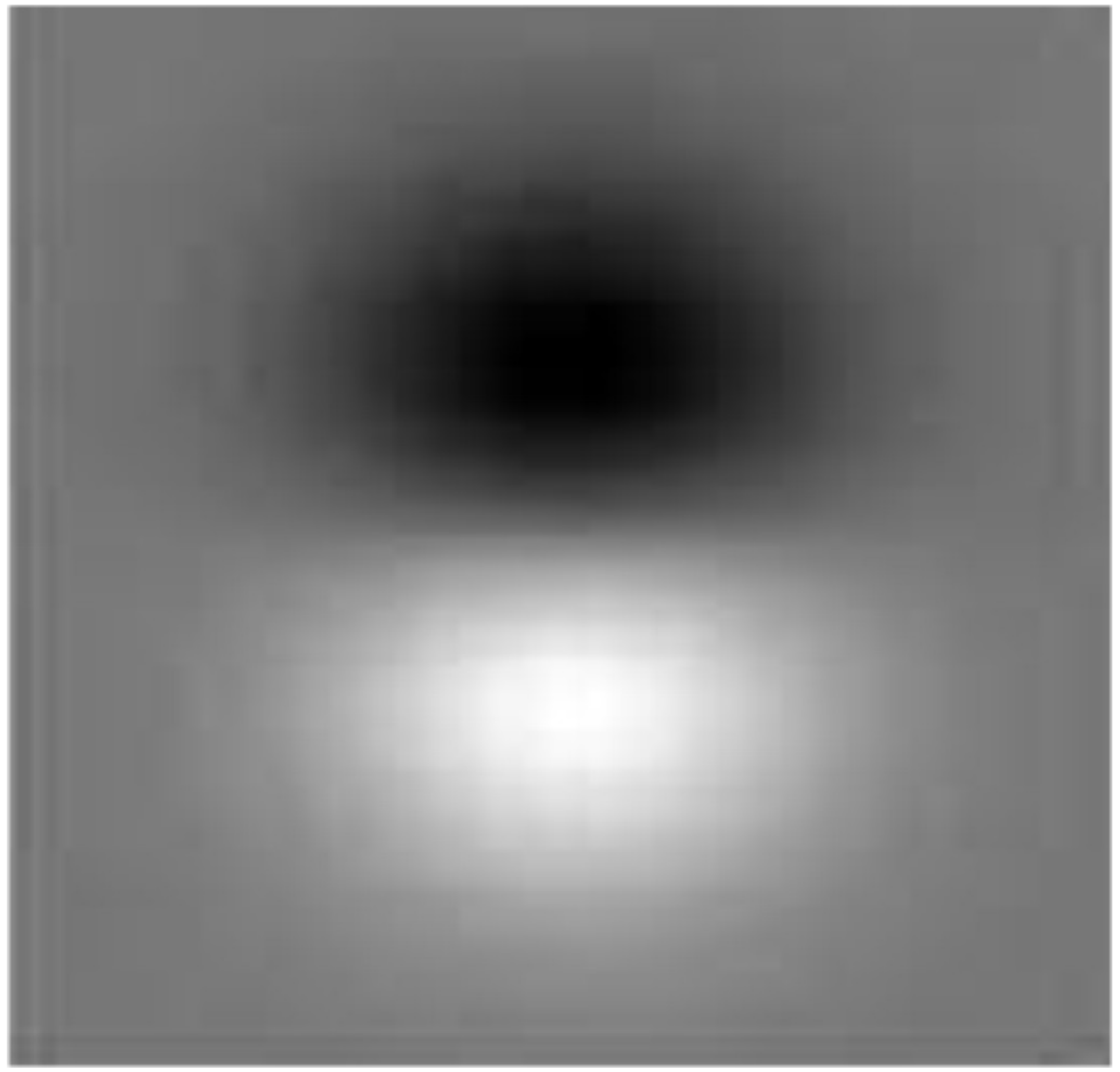
edge

2D Canny Filter

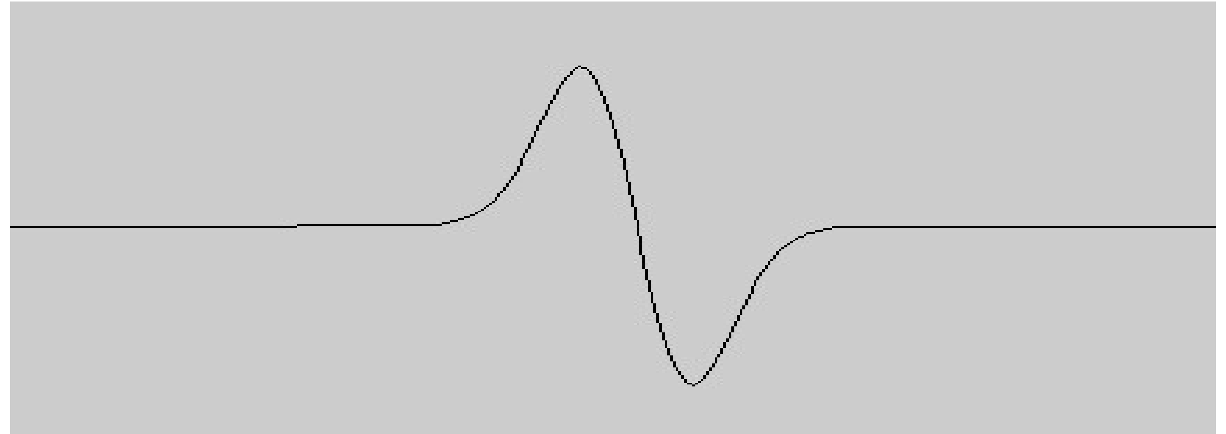
- detection profile orthogonal to edge
- smoothing along edge
  - box filter: Prewitt operator
  - Gauss filter: Sobel operator



detect horizontal edges



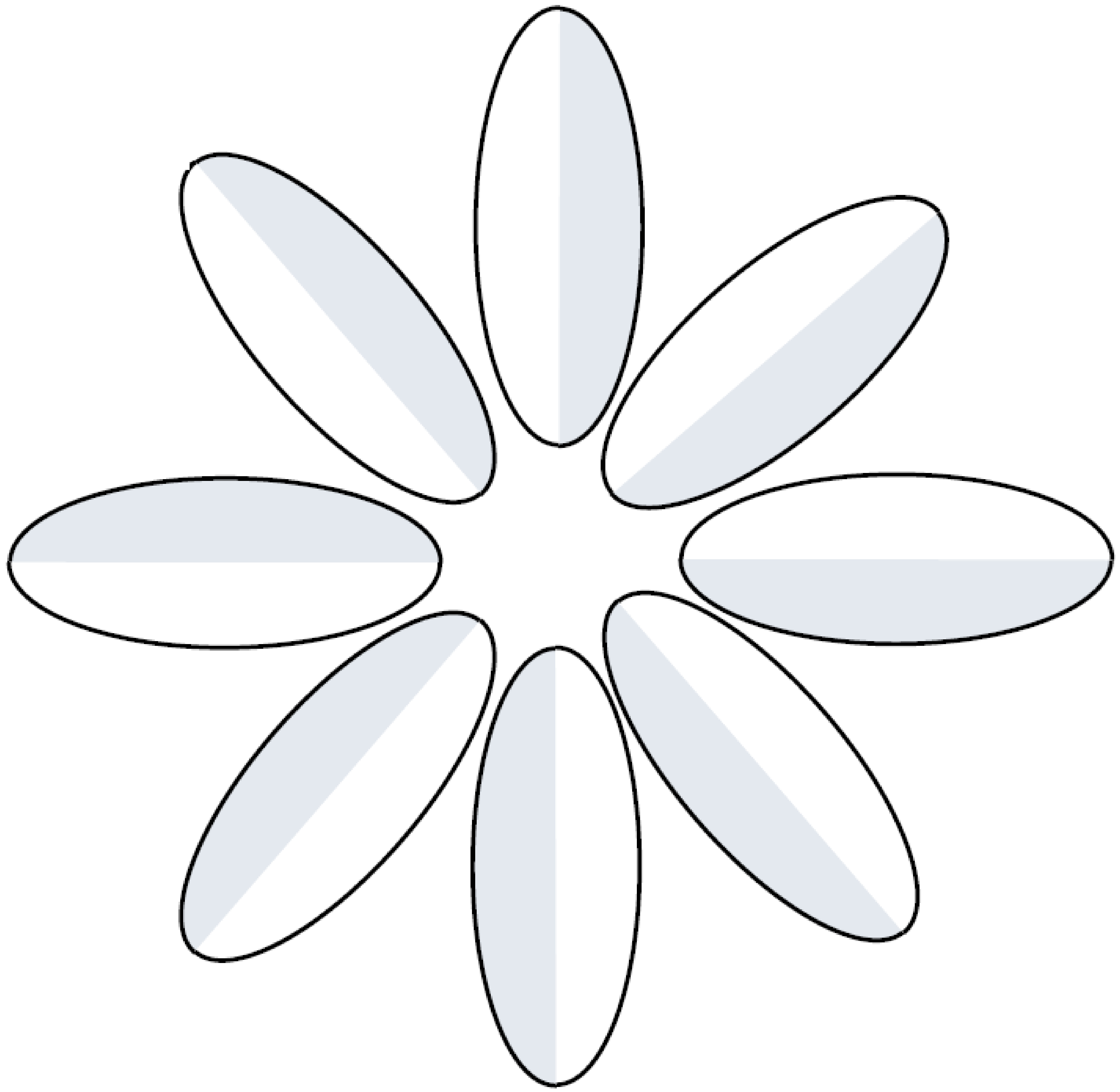
detect vertical edges



profile

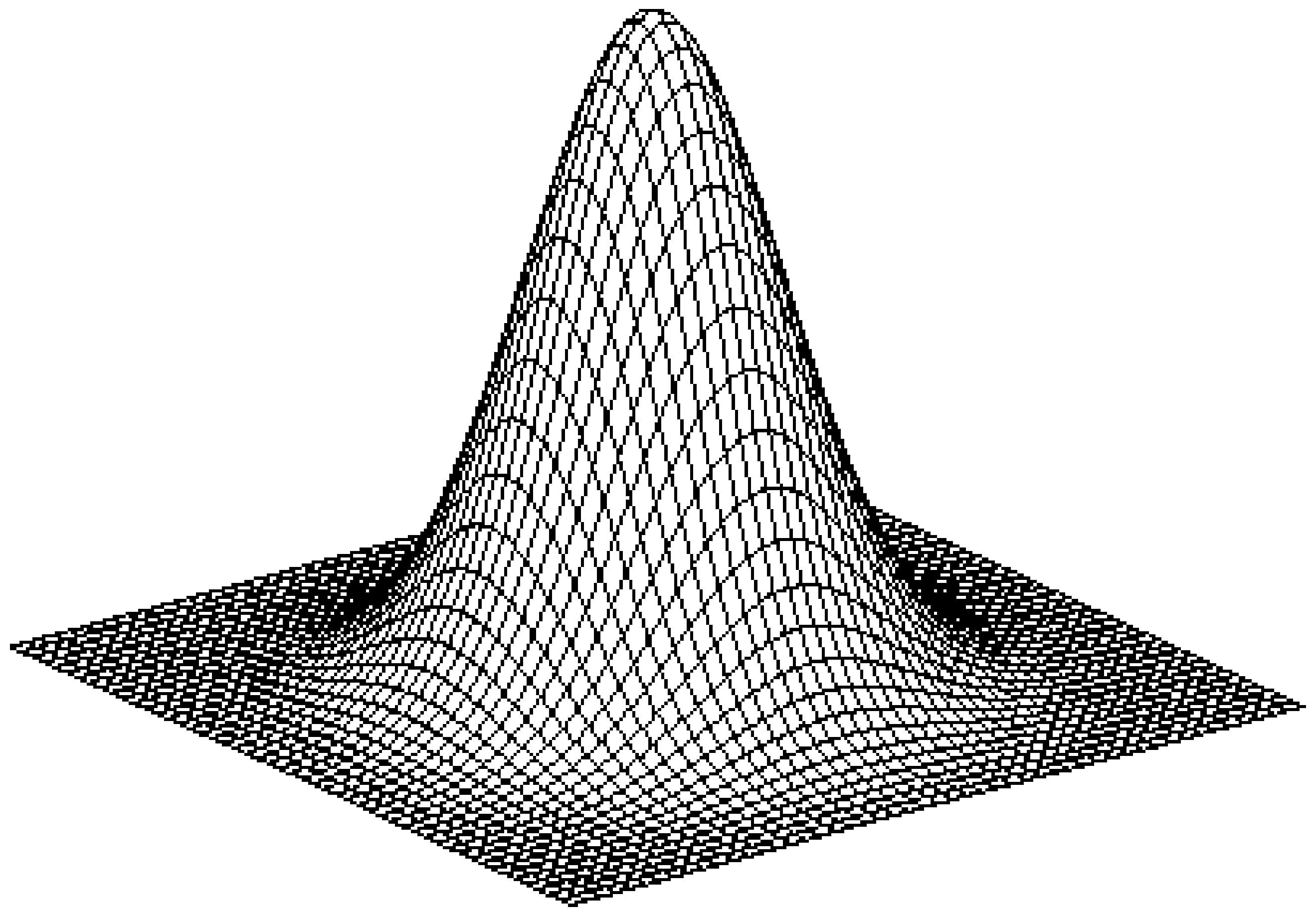
2D Canny Filter

- apply filter in different directions
- look for maximum filter response
- direction of maximum response approximates gradient



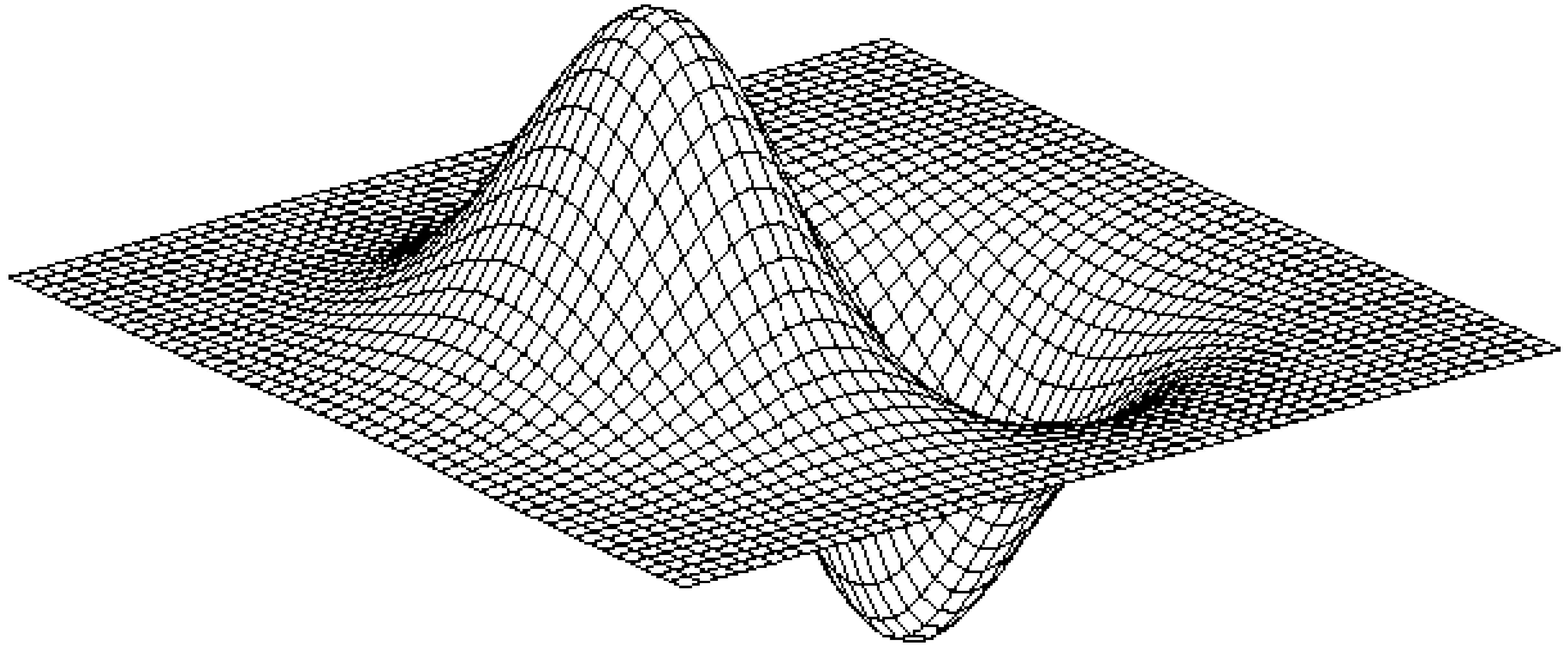
2D Canny Filter

Gauss filter



$$h_\sigma(u,v) \;=\; \frac{1}{2\pi\sigma^2}\,\mathrm{e}^{-\frac{u^2+v^2}{2\sigma^2}}$$

derivative of Gauss filter



$$\frac{\partial}{\partial u}\,h_\sigma(u,v)$$

2D Edge Detection Recipe

Implementation of an isotropic edge detector

- smoothing with 2D Gauss filter
- numerical computation of partial derivatives along x and y axis, dx and dy
- edge strength: sqrt (dx * dx + dy * dy)
- different widths of Gauss filter (sigma) detect finer or more coarse edges (edges in different resolutions)

2D Edge Detection at Different Resolutions

More Edge Detectors

Sobel filter

-1	0	1
-2	0	2
-1	0	1

-1	-2	-1
0	0	0
1	2	1

Prewitt filter

-1	0	1
-1	0	1
-1	0	1

-1	-1	-1
0	0	0
1	1	1

Canny Filter: Additional Steps

Non maxima suppression

- goal: thickness of edges only one pixel
- approach: search for maxima in direction orthogonal to edge

hysteresis thresholding

- use two threshold values
- pixels above higher threshold are edge pixels
- pixels below lower threshold are no edge pixels
- pixels between both threshold values are edge pixels, if they are connected with other edge pixels

Canny Filter: Results